

Deren Lei

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EDUCATION

University of California, Santa Barbara (UCSB)

Computer Science B.S.

College of Engineering Honors Program

Planning to Declare: Minor in Statistical Science

SANTA BARBARA, CA

Fall 2015 – Spring 2019

Major GPA: **3.92** (Around Top 2.5%)

RESEARCH EXPERIENCE

UCSB Natural Language Processing Lab

Researcher (Advisor: William Wang)

SANTA BARBARA, CA

Fall 2017 – Present

- **Stabilized deep reinforcement learning for dialogue (Initiator, work in progress)**
Deep reinforcement learning in NLP have problems of high variance predictions and sparse reward signals. Google's recently released BERT can be a good language modeling prior and solve the above issues.
 - **Inconsistency information solving (Coauthor, in submission)**
Recent entity typing tasks moves to more diverse and fine-grained datasets and current learning approaches have contradiction predictions. Improved performances by avoiding inconsistency outputs using attentions.
 - **Multimodal headline generation (Coauthor, in submission)**
Helping AI powered news content platform company generate attractive smartphone notifications based on contents. Did data collection from CNN news: headline, story highlights, article, images, categories, etc to help our model generates headline via different resources. Our dataset support multiple NLP tasks testing: 1) article classification. 2) headline generation. 3) abstractive summarization.
 - **Implicit regularization of stochastic gradient descent in NLP (First author, in submission)**
Studied the implicit regularization of stochastic gradient descent in practical deep learning systems and its correlation with explicit techniques. Results shed lights on the directions for future deep learning theory studies and also help researchers measure models performances better. ([arXiv:1811.00659](#) [cs.CL])
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WORK EXPERIENCE

Amazon

BEIJING, CHINA

Software Engineering Intern (Mentor: Gang Wang)

Summer 2017

- Created an internal tool to help developers to do automatic product page status checking. Saved them more than 5 hours of manual checking the special offer pages per month.
 - Did more than 100000 data processing to test a softline product recommendation deep learning algorithm that could applied to Amazon product recommendation system in recent years.
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PROJECTS

Activity Recognition (Advisor: B.S. Manjunath)

Winter 2018

- Helping researchers in UCSB Vision Research Lab measuring Residual Computer Networks' performances by learning via single image frames on AVA dataset using ResNet101. The results can help their project of analyzing people's behavior in hospital.

NoteHub (Developer, Product Manager)

Winter 2018

- An software system with android-end and web-end (mainly focused) for students and professors to share notes in both hand-written and electrical format using Java and Javascript.
- Contact Scanbot Team to freely use their API for education purpose when it costs €100,000.

GauchosMap (Group Leader)

Spring 2017

- Developed a map based event sharing system on android platform using Google Maps APIs. To help UCSB students involved in social activities. Get hands-on android development experience with seniors while learning basic engineering skills with other sophomores.
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SKILLS

Programming Languages: Python, Java, C++, Scala, MATLAB, R, MIPS Assembly, Javascript

Technical Tools: Tensorflow, PyTorch, scikit-learn, Git, AWS, LaTeX, SQL, Firebase